

Name: _____

FRACTIONS, DECIMALS AND PERCENTAGES (ORDERING FDP)

Date: _____

LO: I can order fractions, decimals and percentages by converting to a common form.

What does ordering FDP mean?

Ordering FDP means arranging fractions, decimals and percentages from **SMALLEST** to largest (ascending) or largest to smallest (descending).

Strategy: Convert ALL values to the SAME form (usually decimals) before comparing. Then rewrite in original form for your final answer.

Converting to compare

●	$\frac{1}{2}$, 0.6, 55%	0.5, 0.6, 0.55	convert to decimals
●	Order: 0.5, 0.55, 0.6		ascending
●	Answer: $\frac{1}{2}$, 55%, 0.6		rewrite in original form
●	$\frac{1}{3} > \frac{1}{4}$ because $3 < 4$		wrong reasoning
●	$0.9 < 0.12$ because $12 > 9$		ignoring place value

Key idea

Convert everything to decimals. Compare digit by digit from left to right. Write your final answer in the ORIGINAL forms given in the question.

30%, **$\frac{3}{8}$** , **0.34** **0.30, 0.375,**
0.34

Ascending: 30%, 0.34, $\frac{3}{8}$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – two values

Which is larger: $\frac{3}{5}$ or 55%?

$$\frac{3}{5} = 3 \div 5 = 0.6$$

$$55\% = 0.55$$

$$0.6 > 0.55$$

$\frac{3}{5}$ is larger

Convert both to decimals: $0.6 > 0.55$, so $\frac{3}{5}$ is larger.

Example 2 – three values ascending

Order: 0.7, $\frac{3}{4}$, 72%

$$0.7, 0.75, 0.72$$

Smallest first: 0.7

$$0.7, 72\%, \frac{3}{4}$$

0.7, 72%, $\frac{3}{4}$

All as decimals: 0.7, 0.75, 0.72. Ascending: 0.7, 0.72, 0.75.

Example 3 – tricky decimals

Order: 0.3, $\frac{1}{3}$, 30%

$$0.3, 0.333..., 0.30$$

$$0.30 = 0.3 < 0.333...$$

$$30\% = 0.3, \text{ then } \frac{1}{3}$$

0.3 = 30%, then $\frac{1}{3}$

$\frac{1}{3} = 0.333...$ which is greater than 0.3. Note: 0.3 and 30% are equal.

Example 4 – descending order

Order descending: 85%, $\frac{4}{5}$, 0.9

$$0.85, 0.8, 0.9$$

Largest first: 0.9

$$0.9, 85\%, \frac{4}{5}$$

0.9, 85%, $\frac{4}{5}$

Convert: $85\% = 0.85$, $\frac{4}{5} = 0.8$, 0.9. Descending: 0.9, 0.85, 0.8.

YOUR TURN – PRACTICE (deliberate practice)

1. Convert to decimals

Convert each value to a decimal so you can compare them.

- a) $\frac{2}{5}$ = ____
 b) 65% = ____
 c) $\frac{7}{8}$ = ____
 d) 4% = ____

2. Compare two values

Write < or > between each pair. Convert to decimals first.

- a) $\frac{1}{2}$ ____ 45%
 b) 0.3 ____ $\frac{1}{4}$
 c) 68% ____ $\frac{7}{10}$
 d) $\frac{5}{8}$ ____ 60%

3. Order ascending (smallest first)

Write these in ascending order. Show your decimal conversions.

- a) 0.4, $\frac{1}{2}$, 35%
 b) 62%, $\frac{3}{5}$, 0.58
 c) $\frac{3}{8}$, 40%, 0.39
 d) 0.15, 12%, $\frac{1}{8}$

4. Order descending (largest first)

Write these in descending order. Show your decimal conversions.

- a) 0.8, 75%, $\frac{4}{5}$
 b) $\frac{9}{10}$, 88%, 0.92
 c) 55%, 0.6, $\frac{11}{20}$
 d) $\frac{7}{8}$, 90%, 0.86

5. Mixed ordering challenges

Order these sets. State whether ascending or descending as indicated.

- a) Ascending: $\frac{1}{3}$, 30%, 0.29, $\frac{3}{10}$ b) Descending: 0.45, $\frac{9}{20}$, 44%, $\frac{1}{2}$ c) Ascending: 1%, 0.1, $\frac{1}{10}$, 0.01 d) Ascending: $\frac{5}{6}$, 80%, 0.84, $\frac{17}{20}$ e) Descending: 99%, $\frac{19}{20}$, 0.96, 1

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $0.45 > 0.5$ because $45 > 5$ ☐

Correct answer: _____

Reason: _____

b) $\frac{1}{3} > 30\%$ because $\frac{1}{3} = 0.333...$ ☐

Correct answer: _____

Reason: _____

c) $\frac{1}{4} = 14\%$ ☐

Correct answer: _____

Reason: _____

d) $0.08 < 0.1$ ☐

Correct answer: _____

Reason: _____

e) $75\% > \frac{4}{5}$ because $75 > 4$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Five pupils scored these marks on different tests:

Ali: $\frac{17}{25}$ Beth: 70% Cara: 0.68 Dan: $\frac{7}{10}$ Eve: 69%

Rank all five from highest to lowest. Show all conversions.

Expression: _____

Simplified: _____

b) A shop sells three sizes of orange juice:

Small: $\frac{3}{5}$ full for 80p Medium: 58% full for 75p Large: 0.55 full for 70p

Which bottle gives the most juice? Which gives the best value (most juice per penny)?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
 Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

